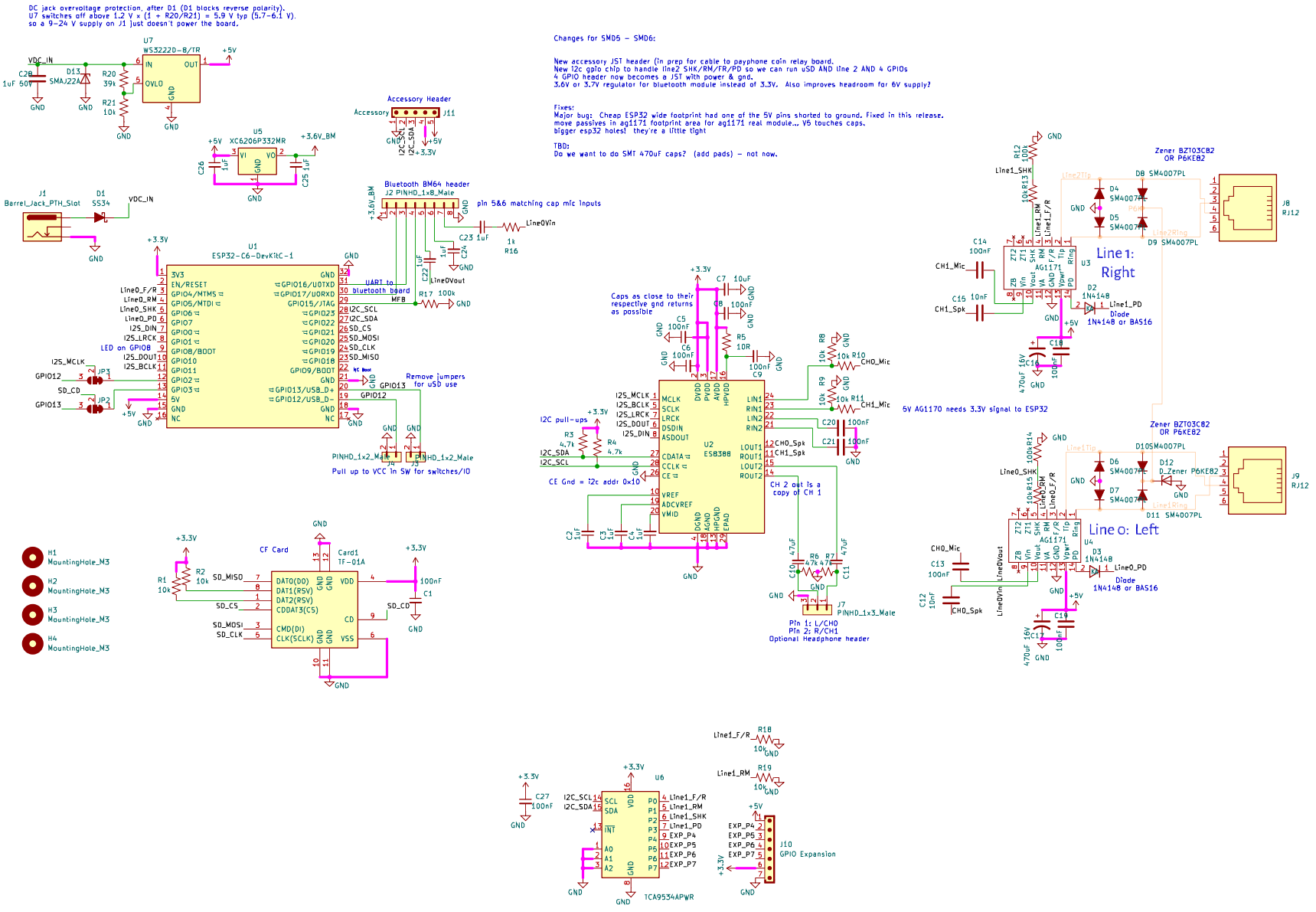

DC jack overvoltage protection, after D1 (D1 blocks reverse polarity).
U7 switches off above 1.2 V × (1 + R20/R21) = 5.9 V typ (5.7–6.1 V).
so a 9–24 V supply on J1 just doesn't power the board.
Changes for SMD5 – SMD6:
New accessory JST header (in prep for cable to payphone coin relay board.
New I2C gpio chip to handle line2_SHK/RM/TR/PD so we can run uSD AND line 2 AND 4 GPIOs
4 GPIO header now becomes a JST with power & gnd.
3.6V or 3.7V regulator for bluetooth module instead of 3.3V. Also improves headroom for 6V supply?
Fixes:
Major bug: Cheap ESP32 wide footprint had one of the 5V pins shorted to ground. Fixed in this release.
move passives in ag1171 footprint area for ag1171 real module.., V5 touches caps.
bigger esp32 holes! they're a little tight
TBD:
Do we want to do SMT 470uF caps? (add pads) – not now.
Accessory Header
Bluetooth BM64 header
pin 5&6 matching cap mic inputs
UART to bluetooth board
Remove jumpers for uSD use
Pull up to VCC in SW for switches/IO
Caps as close to their respective gnd returns as possible
5V AG1170 needs 3.3V signal to ESP32
I2C pull-ups
CE Gnd = I2c addr 0x10
CH 2 out is a copy of CH 1
Optional Headphone header
Zener BZT03CB2 OR P6KE82
Line 1: Right
Line 0: Left
Diode 1N4148 or BAS16
Line 2 + microSD together: U6 (TCA9534A, I2C 0x38) drives line 2's SLIC.
Leave J5 open when U6 is fitted (J5 = line 2 on GPIO18–21 instead of the card).
INT unused; firmware polls SHK, R18/R19 keep F/R and RM low until U6 is set up.
J10: spare expander pins P4–P7.